

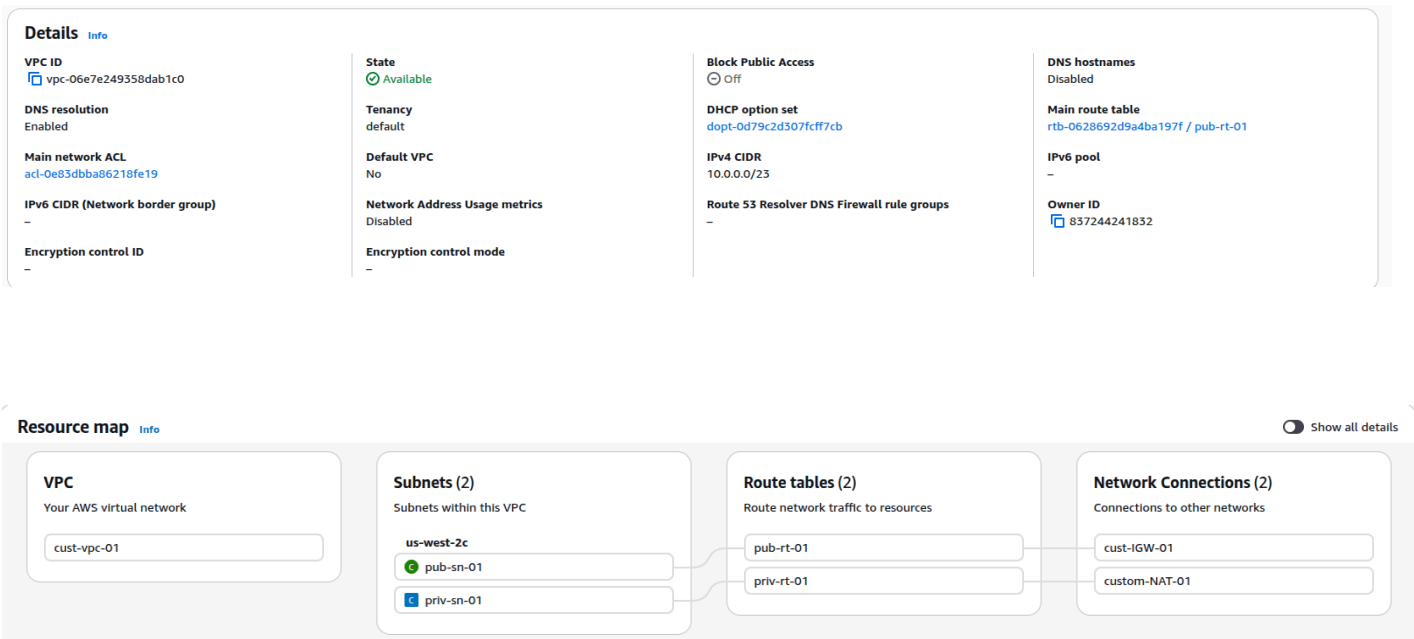
# AWS Task-2

## Task Description:

Set up a VPC with an Internet gateway, create a public subnet with 256 IP addresses, a private subnet with 256 IP addresses, make a route table connecting the Internet gateway and the subnets, and launch a Linux EC2 instance by using the above VPC and public subnet.

## Step 1

Created a custom VPC cust-vpc-01 with CIDR (10.0.0.0/23) 512 IPs



## Step 2 :

Created two subnets

public - With CIDR 10.0.0.0/24 (256 IPs)

**subnet-02fe39ecd52bff9f1 / pub-sn-01** Actions

|                                                      |                                                                                         |                                                   |                                                         |
|------------------------------------------------------|-----------------------------------------------------------------------------------------|---------------------------------------------------|---------------------------------------------------------|
| <b>Details</b>                                       | <b>Subnet ARN</b><br>arn:aws:ec2:us-west-2:837244241832:subnet/subnet-02fe39ecd52bff9f1 | <b>State</b><br>Available                         | <b>Block Public Access</b><br>Off                       |
| <b>Subnet ID</b><br>subnet-02fe39ecd52bff9f1         | <b>Available IPv4 addresses</b><br>251                                                  | <b>IPv6 CIDR</b><br>-                             | <b>IPv6 CIDR association ID</b><br>-                    |
| <b>IPv4 CIDR</b><br>10.0.0.0/24                      | <b>Network border group</b><br>us-west-2                                                | <b>VPC</b><br>vpc-06e7e249358dab1c0   cust-vpc-01 | <b>Route table</b><br>rtb-0628692d9a4ba197f   pub-rt-01 |
| <b>Availability Zone</b><br>usw2-az3 (us-west-2c)    | <b>Default subnet</b><br>No                                                             | <b>Auto-assign public IPv4 address</b><br>No      | <b>Auto-assign IPv6 address</b><br>No                   |
| <b>Network ACL</b><br>acl-0e83dbba86218fe19          | <b>Customer-owned IPv4 pool</b><br>-                                                    | <b>Outpost ID</b><br>-                            | <b>IPv4 CIDR reservations</b><br>-                      |
| <b>Auto-assign customer-owned IPv4 address</b><br>No | <b>IPv6-only</b><br>No                                                                  | <b>Hostname type</b><br>IP name                   | <b>Resource name DNS A record</b><br>Disabled           |
| <b>IPv6 CIDR reservations</b><br>-                   | <b>DNS64</b><br>Disabled                                                                | <b>Owner</b><br>837244241832                      |                                                         |
| <b>Resource name DNS AAAA record</b><br>Disabled     |                                                                                         |                                                   |                                                         |

private - With CIDR 10.0.0.0/24 (256 IPs)

**subnet-077caa723dafee103 / priv-sn-01** Actions

|                                                      |                                                                                         |                                                   |                                                          |
|------------------------------------------------------|-----------------------------------------------------------------------------------------|---------------------------------------------------|----------------------------------------------------------|
| <b>Details</b>                                       | <b>Subnet ARN</b><br>arn:aws:ec2:us-west-2:837244241832:subnet/subnet-077caa723dafee103 | <b>State</b><br>Available                         | <b>Block Public Access</b><br>Off                        |
| <b>Subnet ID</b><br>subnet-077caa723dafee103         | <b>Available IPv4 addresses</b><br>250                                                  | <b>IPv6 CIDR</b><br>-                             | <b>IPv6 CIDR association ID</b><br>-                     |
| <b>IPv4 CIDR</b><br>10.0.1.0/24                      | <b>Network border group</b><br>us-west-2                                                | <b>VPC</b><br>vpc-06e7e249358dab1c0   cust-vpc-01 | <b>Route table</b><br>rtb-0fb1d6a0be7646e29   priv-rt-01 |
| <b>Availability Zone</b><br>usw2-az3 (us-west-2c)    | <b>Default subnet</b><br>No                                                             | <b>Auto-assign public IPv4 address</b><br>No      | <b>Auto-assign IPv6 address</b><br>No                    |
| <b>Network ACL</b><br>acl-0e83dbba86218fe19          | <b>Customer-owned IPv4 pool</b><br>-                                                    | <b>Outpost ID</b><br>-                            | <b>IPv4 CIDR reservations</b><br>-                       |
| <b>Auto-assign customer-owned IPv4 address</b><br>No | <b>IPv6-only</b><br>No                                                                  | <b>Hostname type</b><br>IP name                   | <b>Resource name DNS A record</b><br>Disabled            |
| <b>IPv6 CIDR reservations</b><br>-                   | <b>DNS64</b><br>Disabled                                                                | <b>Owner</b><br>837244241832                      |                                                          |
| <b>Resource name DNS AAAA record</b><br>Disabled     |                                                                                         |                                                   |                                                          |

## Step 3 :

Created internet gateway for public subnet and attached it to custom VPC.

**igw-070cc2bd945862ea3 / cust-IGW-01** Actions

|                                                     |                          |                                                      |                              |
|-----------------------------------------------------|--------------------------|------------------------------------------------------|------------------------------|
| <b>Details</b> <small>info</small>                  | <b>State</b><br>Attached | <b>VPC ID</b><br>vpc-06e7e249358dab1c0   cust-vpc-01 | <b>Owner</b><br>837244241832 |
| <b>Internet gateway ID</b><br>igw-070cc2bd945862ea3 |                          |                                                      |                              |

Step 4 :

Created NAT gateway for custom VPC in public subnet for private subnet.

nat-0e1170861588e7110 / custom-NAT-01

Actions

Details

NAT gateway ID  
nat-0e1170861588e7110

NAT gateway ARN  
arn:aws:ec2:us-west-2:837244241832:natgateway/nat-0e1170861588e7110

VPC  
vpc-06e7e249358dab1c0 / cust-vpc-01

Connectivity type  
Public

Primary public IPv4 address  
54.149.183.234

Subnet  
subnet-077caa723dafee103 / priv-sn-01

State  
Available

Primary private IPv4 address  
10.0.1.92

Created  
Tuesday, February 10, 2026 at 20:28:37 GMT+5:30

State message  
-

Primary network interface ID  
eni-064faadebe65cee7

Deleted  
-

Step 5 :

Created two route tables

Public route table - Added route as destination anywhere and target as Internet gateway.

Routes

Subnet associations

Edge associations

Route propagation

Tags

Routes (2)

Both

Edit routes

Filter routes

| Destination | Target                | Status | Propagated | Route Origin       |
|-------------|-----------------------|--------|------------|--------------------|
| 0.0.0.0/0   | igw-070cc2bd945862ea3 | Active | No         | Create Route       |
| 10.0.0.0/23 | local                 | Active | No         | Create Route Table |

Private route table - Added route as destination anywhere and tagre as NAT gateway

Routes

Subnet associations

Edge associations

Route propagation

Tags

Routes (2)

Both

Edit routes

Filter routes

| Destination | Target                | Status | Propagated | Route Origin       |
|-------------|-----------------------|--------|------------|--------------------|
| 0.0.0.0/0   | nat-0e1170861588e7110 | Active | No         | Create Route       |
| 10.0.0.0/23 | local                 | Active | No         | Create Route Table |

## Step 6 :

Linux EC2 instance by using the custom VPC and public subnet.

The screenshot displays the 'Instance summary' page for the EC2 instance 'webserver02' (ID: i-04f3eb15ce2f75352). The instance is in a 'Running' state. Key details include:

- Instance ID:** i-04f3eb15ce2f75352
- IPv6 address:** -
- Hostname type:** IP name: ip-10-0-0-46.us-west-2.compute.internal
- Answer private resource DNS name:** -
- Auto-assigned IP address:** -
- IAM role:** -
- IMDSv2:** Required
- Operator:** -
- Public IPv4 address:** 54.189.14.72 (with a link to 'open address')
- Instance state:** Running
- Private IP DNS name (IPv4 only):** ip-10-0-0-46.us-west-2.compute.internal
- Instance type:** t3.micro
- VPC ID:** vpc-06e7e249358dab1c0 (cust-vpc-01)
- Subnet ID:** subnet-02fe39ecd52bf9f1 (pub-sn-01)
- Instance ARN:** arn:aws:ec2:us-west-2:837244241832:instance/i-04f3eb15ce2f75352
- Private IPv4 addresses:** 10.0.0.46
- Public DNS:** -
- Elastic IP addresses:** -
- AWS Compute Optimizer finding:** Opt-in to AWS Compute Optimizer for recommendations. (with a link to 'Learn more')
- Auto Scaling Group name:** -
- Managed:** false

## Step 7 :

Added security group to allow ping requests.

The screenshot shows the 'Details' and 'Inbound rules' sections of a security group.

**Details:**

- Security group name:** SSH
- Owner:** 837244241832
- Security group ID:** sg-0f06c312aab98b761
- Inbound rules count:** 1 Permission entry
- Description:** SSH
- Outbound rules count:** 1 Permission entry
- VPC ID:** vpc-06e7e249358dab1c0

**Inbound rules (1):**

| Name | Security group rule ID | IP version | Type            | Protocol | Port range | Source    | Description |
|------|------------------------|------------|-----------------|----------|------------|-----------|-------------|
| -    | sg-086a0bd0acd1ecf52   | IPv4       | All ICMP - IPv4 | ICMP     | All        | 0.0.0.0/0 | -           |

And server is publicly accessible.

The screenshot shows an 'Online service Ping' tool interface. The IP address entered is 54.189.14.72. The tool shows successful ping results with a response time of 167 ms for each of the four packets sent.

--- PING 54.189.14.72 (54.189.14.72) 56(84) bytes of data: ---  
64 bytes from 54.189.14.72: icmp\_seq=1 ttl=115 time=167 ms  
64 bytes from 54.189.14.72: icmp\_seq=2 ttl=115 time=167 ms  
64 bytes from 54.189.14.72: icmp\_seq=3 ttl=115 time=167 ms  
64 bytes from 54.189.14.72: icmp\_seq=4 ttl=115 time=167 ms

--- 54.189.14.72 ping statistics ---  
packets transmitted 4  
received 4  
packet loss 0 %  
time 3014 ms

